Predictive Analytics for Smart City Infrastructure

M.Tech Project Work Final Review

Presented by

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(Artificial Intelligence and Machine Learning – In collaboration with LTI Mindtree)

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1.ABSTRACT

Traffic control rooms can see congestion as it happens, but rarely receive forecasts that are explained, honest about their uncertainty and tested before an action is suggested. This project builds a decision-support prototype on public New York City data. A context-gated GATv2–Transformer forecasts link speeds 15, 30 and 60 minutes ahead on a graph of 16 Manhattan DOT speed links (January–October 2026), fusing weather, 311 and calendar context through an availability-aware gate. Split-conformal intervals quantify uncertainty, grouped SHAP explains the trained model, and a LangGraph workflow uses a local LLM (qwen2.5 7B) with FAISS retrieval to draft signal advisories that deterministic safety rules and a SUMO simulation must pass before human review. On a 45-day test period the model reaches 4.77, 5.33 and 6.15 mph MAE, 7–16 % below persistence; intervals reach 90 % coverage overall, and retrieval removed every fabricated citation. On a calibrated SUMO testbed (10.08 vs 10.95 mph observed) a forecast-driven rule cut simulated time loss by 5.4 % and 4.7 % in surge and incident scenarios, a simulated result only.

2. INTRODUCTION

- Urban congestion depends on demand, neighbouring roads, weather, events and incidents, yet control rooms mostly see only current conditions.
- Graph neural networks forecast traffic on road networks [1]–[3], but give point forecasts without uncertainty or explanation.
- An LLM can phrase advice convincingly without evidence; retrieval grounding [4] and simulation testing [5] are needed before any signal action.
- Objective: join forecasts, uncertainty, explanations, cited evidence and simulation checks in one auditable, human-in-the-loop advisory pipeline on real NYC data.

[1] Y. Li, R. Yu, C. Shahabi, and Y. Liu, “Diffusion convolutional recurrent neural network: Data-driven traffic forecasting,” in Proc. ICLR, 2018.
[2] B. Yu, H. Yin, and Z. Zhu, “Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting,” in Proc. IJCAI, 2018, pp. 3634–3640.
[3] Z. Wu, S. Pan, G. Long, J. Jiang, and C. Zhang, “Graph WaveNet for deep spatial-temporal graph modeling,” in Proc. IJCAI, 2019, pp. 1907–1913.
[4] P. Lewis et al., “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in Proc. NeurIPS, 2020.
[5] P. A. Lopez et al., “Microscopic traffic simulation using SUMO,” in Proc. IEEE ITSC, 2018, pp. 2575–2582.

3.LITERATURE SURVEY

S.No	Article Title	Techniques Used	Key Findings	Limitations
1	[1] Li et al., Diffusion Convolutional Recurrent Neural Network (ICLR 2018)	Diffusion convolution on a directed road graph inside a GRU encoder–decoder (seq2seq)	Modelling traffic as diffusion on the directed graph improved multi-step speed forecasts over non-graph baselines	Fixed, predefined graph; point forecasts only; no exogenous context or uncertainty
2	[2] Yu et al., Spatio-Temporal Graph Convolutional Networks (IJCAI 2018)	Graph convolution combined with gated temporal convolution; fully convolutional, no recurrence	Accurate short-term forecasts with faster training and fewer parameters than recurrent models	Static graph; point forecasts; no weather or event context
3	[3] Wu et al., Graph WaveNet (IJCAI 2019)	Self-adaptive adjacency matrix learned from data plus dilated causal convolutions	Captures hidden spatial dependencies even when the given graph is incomplete; long receptive field	Learned graph is hard to interpret; no uncertainty estimates
4	[6] Brody et al., How Attentive are Graph Attention Networks? (ICLR 2022)	GATv2: reorders the attention computation so that attention becomes dynamic	Shows that GAT attention is static; GATv2 is strictly more expressive and performs better on benchmarks	General-purpose graph layer; no temporal modelling

[1] Y. Li, R. Yu, C. Shahabi, and Y. Liu, “Diffusion convolutional recurrent neural network: Data-driven traffic forecasting,” in Proc. ICLR, 2018.
[2] B. Yu, H. Yin, and Z. Zhu, “Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting,” in Proc. IJCAI, 2018, pp. 3634–3640.
[3] Z. Wu, S. Pan, G. Long, J. Jiang, and C. Zhang, “Graph WaveNet for deep spatial-temporal graph modeling,” in Proc. IJCAI, 2019, pp. 1907–1913.
[6] S. Brody, U. Alon, and E. Yahav, “How attentive are graph attention networks?,” in Proc. ICLR, 2022.

3.LITERATURE SURVEY (CONTD.)

S.No	Article Title	Techniques Used	Key Findings	Limitations
5	[7] Angelopoulos & Bates, A Gentle Introduction to Conformal Prediction (2021)	Split conformal prediction wrapped around any trained model	Distribution-free, finite-sample marginal coverage guarantee	Assumes exchangeable data; guarantees marginal, not conditional, coverage
6	[8] Xu & Xie, Conformal Prediction Interval for Dynamic Time-Series (ICML 2021)	EnbPI: bootstrap ensemble with conformal-style intervals for time series	Approximately valid intervals without exchangeability, under stated assumptions on the errors	Requires an ensemble of models; validity is approximate
7	[9] Lundberg & Lee, A Unified Approach to Interpreting Model Predictions (NeurIPS 2017)	SHAP: Shapley-value feature attribution (KernelSHAP and model-specific variants)	Unifies additive attribution methods with consistency guarantees	Cost grows with the number of features; attributions describe the model, not causes
8	[4] Lewis et al., Retrieval-Augmented Generation (NeurIPS 2020)	Dense passage retriever combined with a seq2seq generator	More specific and factual generation on knowledge-intensive tasks than a parametric-only model	Bounded by corpus and retriever quality; can still produce unsupported text

[4] P. Lewis et al., “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in Proc. NeurIPS, 2020.
[7] A. N. Angelopoulos and S. Bates, “A gentle introduction to conformal prediction and distribution-free uncertainty quantification,” arXiv:2107.07511, 2021.
[8] C. Xu and Y. Xie, “Conformal prediction interval for dynamic time-series,” in Proc. ICML, PMLR vol. 139, 2021, pp. 11559–11569.
[9] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in Proc. NeurIPS, 2017.

4. RESEARCH GAPS

G1 Forecast stops at the number

Graph forecasting studies [1]–[3] report accuracy but do not connect the forecast to a checked, testable action.

G2 Uncertainty does not decide

Point forecasts dominate; interval width and data quality rarely decide whether an advisory is issued.

G3 Surrogate explanations

Explanations are often computed on simpler surrogate models instead of the deployed forecaster.

G4 Ungrounded LLM advice

LLM recommendations are rarely tied to cited, allow-listed evidence or bounded by deterministic safety rules.

G5 Missing context is hidden

Dead sensors and unavailable feeds are usually imputed silently instead of being modelled with availability masks.

G6 Untested testbeds

Simulation studies seldom audit route choice, demand and signal timing before comparing controllers.

5. OBJECTIVES

O1 Forecast link speeds 15, 30 and 60 min ahead with a context-gated GATv2–Transformer and compare it with six baselines on identical data

Addresses: G1, G5

O2 Quantify uncertainty with per-horizon split-conformal intervals and defer advisories when reliability is low

Addresses: G2

O3 Explain the trained model itself with grouped SHAP and sensitivity tests

Addresses: G3

O4 Draft advisories with a local LLM that must cite retrieved evidence and pass deterministic safety checks

Addresses: G4

O5 Test admissible advisories in a SUMO simulation, after verifying the testbed itself

Addresses: G6

O6 Make every result reproducible and traceable: tests, MLflow, Docker, evidence registry and a paper generated from artifacts

Addresses: all

6. RESEARCH WORK DONE

- The research work is organised in four parts:
  - ❖ **MODULE IDENTIFICATION**
  - ❖ **SYSTEM ARCHITECTURE**
  - ❖ **DEVELOPMENT & INTEGRATION**
  - ❖ **PROTOTYPE / RESEARCH FRAMEWORK**

1,968,303
raw speed records processed

16
Manhattan links modelled

7
forecasting models compared

112
automated tests passing

MODULE IDENTIFICATION

M1 Data ingestion
NYC Open Data, Open-Meteo, OSM; resumable downloads with SHA-256 provenance

M2 Preprocessing & features
5-minute grid, causal lags, availability masks, leakage-tested splits

M3 Road graph
Directed DOT-link graph (26 nodes, 16 edges) as a PyG Temporal signal [10]

M4 Forecasting
Context-gated GATv2–Transformer plus six baselines at 15/30/60 min

M5 Uncertainty
Per-horizon split-conformal intervals and a reliability gate

M6 Explainability
Grouped KernelSHAP on the trained model; neighbour and window sensitivity

M7 Knowledge & RAG
15 approved documents, MiniLM embeddings, FAISS index

M8 LLM agent
LangGraph workflow; local qwen2.5 7B via Ollama; schema-validated JSON

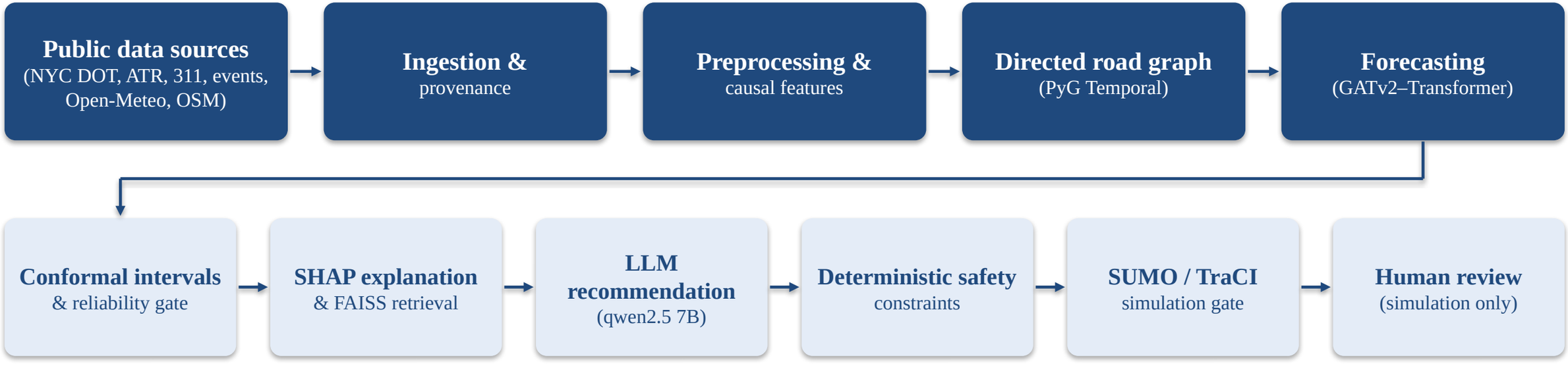
M9 Safety constraints
13 deterministic checks; protected TraCI signal controller

M10 Simulation
SUMO 1.28 West Street corridor; matched seeds; testbed audit

M11 Services
FastAPI, PostgreSQL/PostGIS job queue, worker, Streamlit dashboard

M12 Tracking & reporting
MLflow, evidence registry, claims ledger, paper tables from artifacts

SYSTEM ARCHITECTURE



Data flow

Raw API responses stored verbatim with hashes → 5-minute grid per link → windows of 24 steps × nodes × features → forecasts with 90 % intervals → advisory state in LangGraph → decision and audit trail persisted in PostgreSQL/PostGIS; every run logged in MLflow.

Technology stack

Python 3.12 · Pandas/Polars · PyTorch + PyTorch Geometric + PyG Temporal · XGBoost/LightGBM · SHAP · sentence-transformers + FAISS · Ollama (qwen2.5:7b) · LangGraph · SUMO 1.28 / TraCI · FastAPI · Streamlit · PostgreSQL/PostGIS · MLflow · Docker Compose · Pytest

DEVELOPMENT & INTEGRATION

Implementation of modules

One Python package with a staged command-line interface, YAML configuration and Pydantic contracts

Every stage writes artifacts with run IDs and SHA-256 hashes

The same code path runs the smoke tests and the full experiment

Integration strategy

Shared contracts frozen first: forecast, uncertainty, evidence, candidate action, decision

A stage runner executes the real-data pipeline in fixed order with a stage log

Only the recommendation node calls the LLM; every gate is deterministic code

112 automated tests: unit, leakage, safety, integration, end-to-end (CI, 2026-10-28)

Challenges faced → solutions

7 of 26 links report zero speed all year
→ treated as missing; 16 links kept by a train-coverage rule

DOT link geometries stop short of link ends
→ three documented graph-edge rules

GRU crash with cuDNN on Windows/CUDA
→ cuDNN disabled for GRU runs only

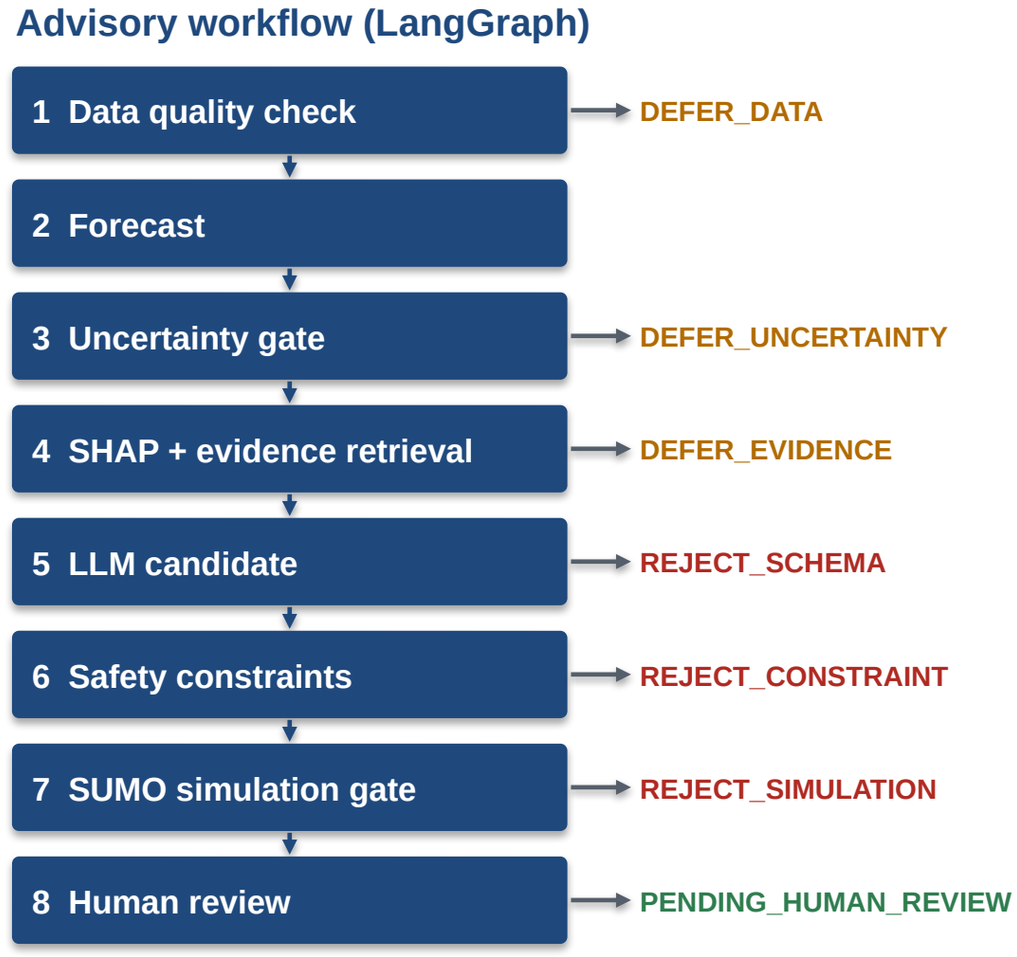
Bootstrap resampled 5-minute steps, not days
→ unit fixed and ablations re-run

A missing field silently blocked every LLM call
→ fixed and agent evaluation re-run

SUMO re-routed half the corridor traffic
→ fixed routes and a testbed audit

PROTOTYPE / RESEARCH FRAMEWORK

- **Command-line pipeline:** one command per stage, from data audit to paper tables
- **FastAPI service:** 20 endpoints for forecasts, live forecasts, explanations, retrieval, advisories, simulations and review; long jobs run in a PostgreSQL-backed worker
- **Streamlit dashboard:** 12 pages, from data audit, traffic map and live monitoring to forecasts, explanations, agent traces, SUMO results and the model card
- **Every view labels its mode:** LIVE, REPLAY or SIMULATION
- **Docker Compose:** PostGIS, API, worker and dashboard images, built and smoke-tested
- **Human review** approves or rejects for simulation only; nothing is sent to real signals



7. CONCLUSIONS

Performance metrics (test period 6 Sep – 20 Oct 2026)

Metric	15 min	30 min	60 min
Proposed model MAE (mph)	**4.77**	**5.33**	**6.15**
Proposed, mean of 3 seeds (mph)	4.69	5.34	6.15
Persistence MAE (mph)	5.14	6.06	7.29
90 % interval coverage	0.909	0.907	0.903
Coverage under congestion	0.798	0.794	0.757
MAE kept after gate (≈26 % deferred)	3.72	4.38	5.25

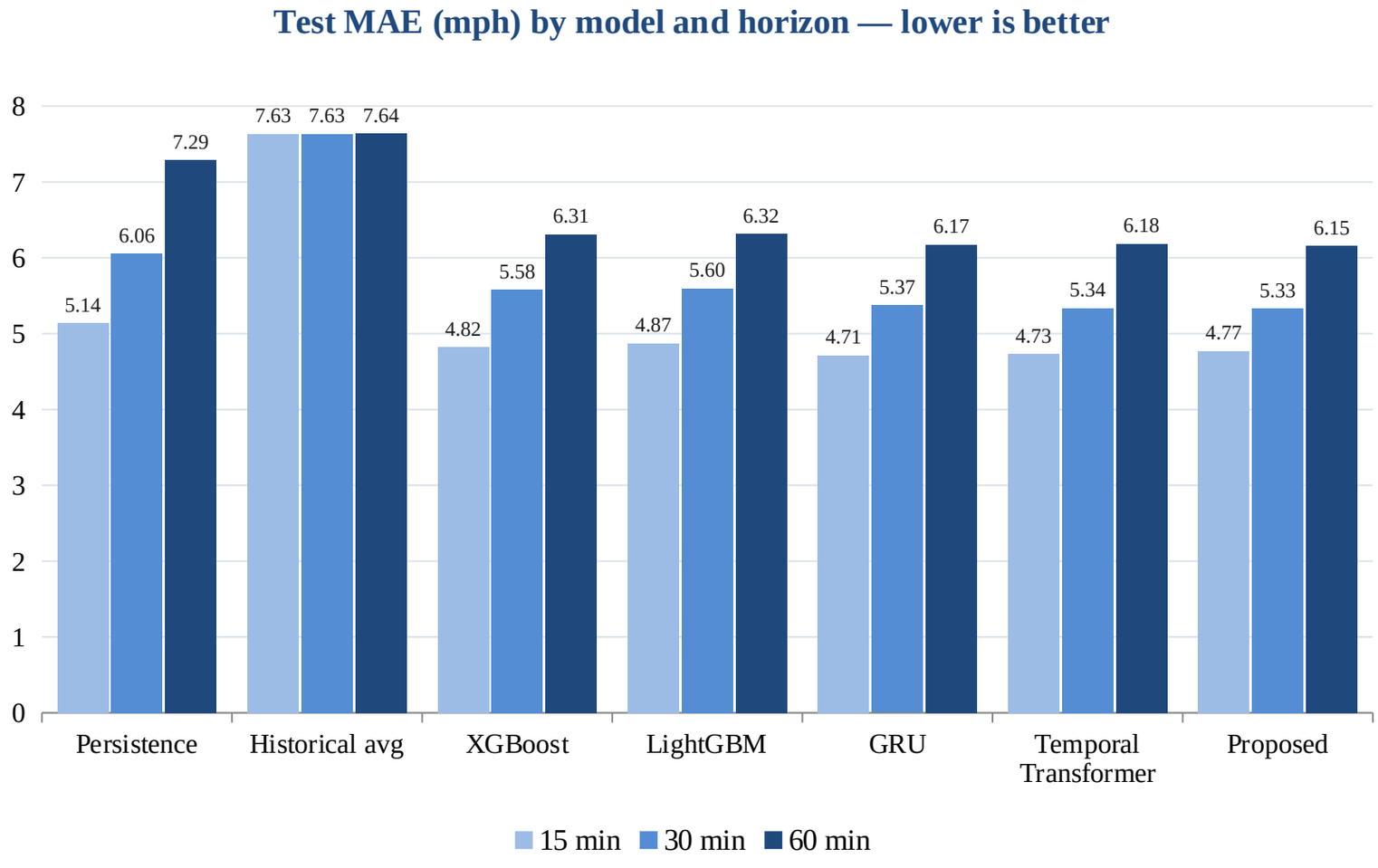
Validation of approach

Chronological splits with purge gaps and leakage tests; identical targets and masks for every model; day-block bootstrap for ablations; a preregistered simulation baseline test; failed runs kept and reported.

Observations

- The proposed model beats persistence by 7–16 % and both tree models. Over three seeds it is 0.03–0.08 mph below the temporal Transformer at every horizon and the GRU at 30/60 min; the GRU is better at 15 min. Small, not decisive.
- The availability-aware gate helps (MAE +0.067 mph when removed); operational weather features hurt (−0.046 mph), a negative result.
- Intervals meet 90 % overall but under-cover congestion, the regime that matters for advisories.
- Retrieval removed fabricated citations (3 of 3 without it, 0 of 9 with it); the constraint layer, not the LLM, decides admissibility.
- On the calibrated SUMO testbed the forecast rule lowered simulated time loss by 5.4 % (surge) and 4.7 % (incident) against fixed time in every seed; routine peak unchanged. Simulated, not a real-world gain.

8. OUTCOME



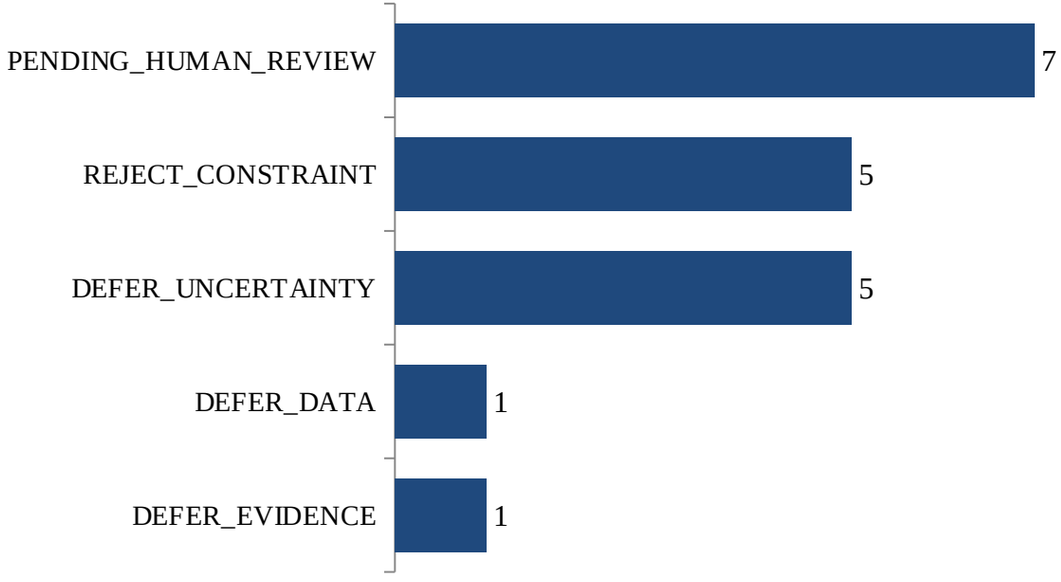
Key results

- Every learned model beats persistence and the historical average.
- Proposed model: 4.77 / 5.33 / 6.15 mph; lowest at 30 and 60 min, GRU lowest at 15 min.
- R^2 rises from 0.55 / 0.46 / 0.30 (persistence) to 0.69 / 0.63 / 0.56.
- 144,205–144,217 test targets per horizon, identical for every model.

Data: NYC DOT speeds [11], 1 Jan – 20 Oct 2026.

8. OUTCOME (CONTD.)

Advisory decisions in 19 workflow runs (REPLAY cases)



12 of 12 LLM responses were schema-valid. Without retrieval the model cited non-existent evidence in 3 of 3 cases (all rejected); with retrieval in 0 of 9.

SUMO testbed calibration (SIMULATION)

Protocol	Unfinished*	Result
First comparison, 20 Oct (45 runs)	—	**Superseded**
2014 / 2025 counts, generated or Webster timing	4.3–17.7 %	**Fails test**
v5: corridor green ×1.0–2.5	0–0.5 % at ×2.0+	**8.1 of 10.95 mph**
v6: 90 s cycle, green ×1.5, 25 mph offsets	0 %	**10.08 mph: passes**

* Vehicles not finishing with no side-street traffic; the preregistered test allows at most 1 % and no stuck vehicles.

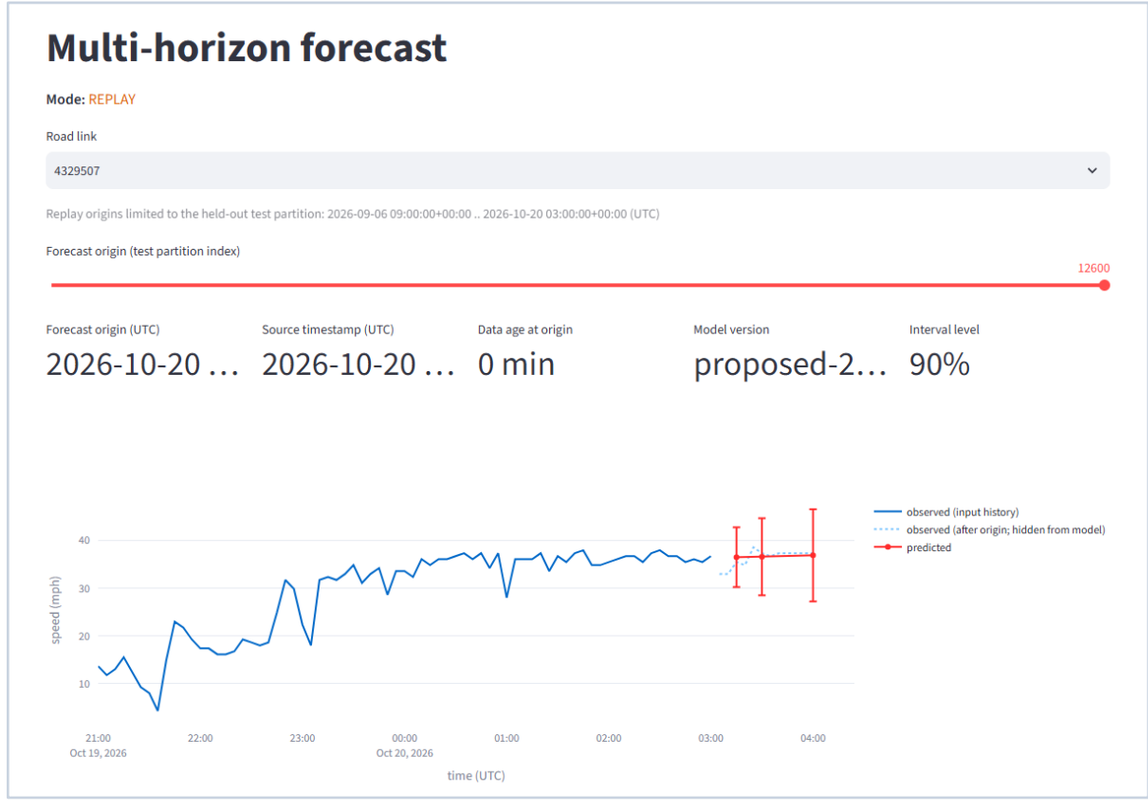
The first comparison let SUMO re-route 1,120 of 2,189 main-road vehicles off West Street, so it did not test the corridor.

Target: the real PM-peak speed of 10.95 mph (±1.5) with 2025 counts. Splits alone peaked at 8.88 mph; southbound offsets closed the gap; 120 s cells overshot (13–22 mph).

v6 (26 Oct, 75 runs) selects 90 s / ×1.5 / 25 mph: 10.08 mph, 0 unfinished. A calibrated scenario testbed, not NYC DOT timing; only side-street demand 0.0 is admissible.

8. OUTCOME: WORKING DASHBOARD (1/5)

Multi-horizon forecast (proposed model) Mode shown: REPLAY



Traffic map: monitored links Mode shown: REPLAY



Captured on 23 Oct 2026 on the training laptop, which holds the trained model and the processed NYC DOT data. The forecast is a REPLAY of the held-out test period (origin 20 Oct 2026, 90 % conformal intervals); the LIVE page is on slide 23.

8. OUTCOME: WORKING DASHBOARD (2/5)

Overview & status (status page, no mode label)

Multi-horizon traffic forecasting & explainable agentic signal advisory

Predictive Analytics for Smart City Infrastructure - Multi-Horizon Traffic Congestion Forecasting and Explainable Agentic Signal Advisory Using Spatiotemporal Transformers and RAG

Proposed model	Track	Study area
proposed-20261020T…	operational	manhattan_dot_spee…

Component status (evidence registry)

requirement_id	component	test_status	artifact_path	limitations
DATA-01	data	VERIFIED	reports/data_audit/source_inventory.csv	None
DATA-02	data	PARTIAL	data/manifests/source_manifest.json	The forecasting data (sp
DATA-03	data	VERIFIED	reports/data_audit/availability_audit.md	None
GRAPH-01	graph	VERIFIED	reports/data_audit/graph_validation.json	None
GRAPH-02	graph	VERIFIED	artifacts/models/proposed-20261020T210211-4a0685/feature_schema.json	None
MODEL-01	models	VERIFIED	artifacts/models/registry_operational.json	None
MODEL-02	models	VERIFIED	reports/experiments/forecast_metrics.csv	None
UNC-01	uncertainty	VERIFIED	artifacts/calibration/calibrator.json	None
XAI-01	explainability	VERIFIED	artifacts/explanations/explanations_summary.json	None
RAG-01	rag	VERIFIED	artifacts/retrieval/retrieval_eval.json	None

› final_status.md

Data audit & provenance Mode shown: REPLAY

Data audit & provenance

Mode: REPLAY

source_id	official_title	publisher	dataset_id	endpoint
speeds	DOT Traffic Speeds	City of New York	i4gi-tjb9	https://data.cityofnewyork.us/resource/i4gi-tjb9.json
volume	Automated Traffic Volume Counts	City of New York	7ym2-wayt	https://data.cityofnewyork.us/resource/7ym2-wayt.json
events	NYC Permitted Event Information	Office of Citywide Event Coord	tvpp-9vvx	https://data.cityofnewyork.us/resource/tvpp-9vvx.json
events_hist	NYC Permitted Event Information - Historical	City of New York	bkfu-528j	https://data.cityofnewyork.us/resource/bkfu-528j.json
collisions	Motor Vehicle Collisions - Crashes	Police Department (NYPD)	h9gi-nx95	https://data.cityofnewyork.us/resource/h9gi-nx95.json
sr311	311 Service Requests from 2020 to Present	311	erm2-nwe9	https://data.cityofnewyork.us/resource/erm2-nwe9.json
weather	Open-Meteo archive (ERA5-based reanalysis) + historical forecast API	Open-Meteo		https://archive-api.open-meteo.com/v1/archive;https://historic
osm	OpenStreetMap extracts via Overpass API	OpenStreetMap contributors		https://overpass-api.de/api/interpreter

Coverage by year (server-side counts)

source_id	dataset_id	year	record_count
speeds	i4gi-tjb9	1930	8
speeds	i4gi-tjb9	1978	15
speeds	i4gi-tjb9	2014	6
speeds	i4gi-tjb9	2015	9
speeds	i4gi-tjb9	2016	36
speeds	i4gi-tjb9	2017	6019989
speeds	i4gi-tjb9	2018	13860107
speeds	i4gi-tjb9	2019	12971855
speeds	i4gi-tjb9	2020	11943235
speeds	i4gi-tjb9	2021	12203801

Screenshots of the project's Streamlit dashboard (src/smartcity/dashboard/app.py) captured on 28 Oct 2026; every number on them is read from committed result files. The LIVE page, run on 30 Oct 2026, is on slide 23.

8. OUTCOME: WORKING DASHBOARD (3/5)

Experiment results Mode shown: REPLAY

Experiment results

Mode: REPLAY

forecast_metrics

run_id	model	track	split	evaluation_period_utc	horizon_min	n_valid_targets	mae_mp
persistence-20261020T193923-c48849	persistence	operational	test	2026-09-06 09:00:00+00:00 .. 2026-10-20 04:00:00+00:00	15	144217	5.138604
persistence-20261020T193923-c48849	persistence	operational	test	2026-09-06 09:00:00+00:00 .. 2026-10-20 04:00:00+00:00	30	144213	6.055625
persistence-20261020T193923-c48849	persistence	operational	test	2026-09-06 09:00:00+00:00 .. 2026-10-20 04:00:00+00:00	60	144205	7.287564
historical_average-20261020T193932-e08975	historical_average	operational	test	2026-09-06 09:00:00+00:00 .. 2026-10-20 04:00:00+00:00	15	144217	7.632188
historical_average-20261020T193932-e08975	historical_average	operational	test	2026-09-06 09:00:00+00:00 .. 2026-10-20 04:00:00+00:00	30	144213	7.632955
historical_average-20261020T193932-e08975	historical_average	operational	test	2026-09-06 09:00:00+00:00 .. 2026-10-20 04:00:00+00:00	60	144205	7.636957
xgboost-20261020T193933-18d124	xgboost	operational	test	2026-09-06 09:00:00+00:00 .. 2026-10-20 04:00:00+00:00	15	144217	4.822946
xgboost-20261020T193933-18d124	xgboost	operational	test	2026-09-06 09:00:00+00:00 .. 2026-10-20 04:00:00+00:00	30	144213	5.579227
xgboost-20261020T193933-18d124	xgboost	operational	test	2026-09-06 09:00:00+00:00 .. 2026-10-20 04:00:00+00:00	60	144205	6.307995
lightgbm-20261020T193956-ccbf73	lightgbm	operational	test	2026-09-06 09:00:00+00:00 .. 2026-10-20 04:00:00+00:00	15	144217	4.872539

ablation_metrics

variant	track	run_id	mae_mean_mph	delta_vs_proposed_mph	ci95_low	ci95_high	n_days	status
proposed	operational	proposed-20261020T201110-0e333f	5.4194	0.0	0.0	0.0	45	REFERENC
ablation_no_gate	operational	ablation_no_gate-20261020T202053-48c9ff	5.4867	0.0673	0.0482	0.0864	44	OK
ablation_no_weather	operational	ablation_no_weather-20261020T202610-295c01	5.3732	-0.0462	-0.0692	-0.0205	44	OK
ablation_no_infrastructure	operational	ablation_no_infrastructure-20261020T203545-6cd844	5.433	0.0136	0.0019	0.0258	44	OK
ablation_no_events	operational	ablation_no_events-20261020T204114-372604						NOT_APPL

Uncertainty & reliability Mode shown: REPLAY

coverage_by_horizon.csv

horizon_min	group	n	coverage	mean_width_mph
15	all	144217	0.909	24.2562
30	all	144213	0.9074	26.945
60	all	144205	0.9033	29.5394

coverage_by_scenario.csv

horizon_min	group	n	coverage	mean_width_mph
15	free_ratio>=0.8	98556	0.9311	24.2562
15	moderate_0.5-0.8	22396	0.9266	24.2562
15	congested<0.5	23265	0.7982	24.2562
30	free_ratio>=0.8	98546	0.9301	26.945
30	moderate_0.5-0.8	22398	0.9259	26.945
30	congested<0.5	23269	0.7935	26.945
60	free_ratio>=0.8	98522	0.9329	29.5394
60	moderate_0.5-0.8	22406	0.9253	29.5394
60	congested<0.5	23277	0.7569	29.5394
15	non_rush	106048	0.8984	24.2562

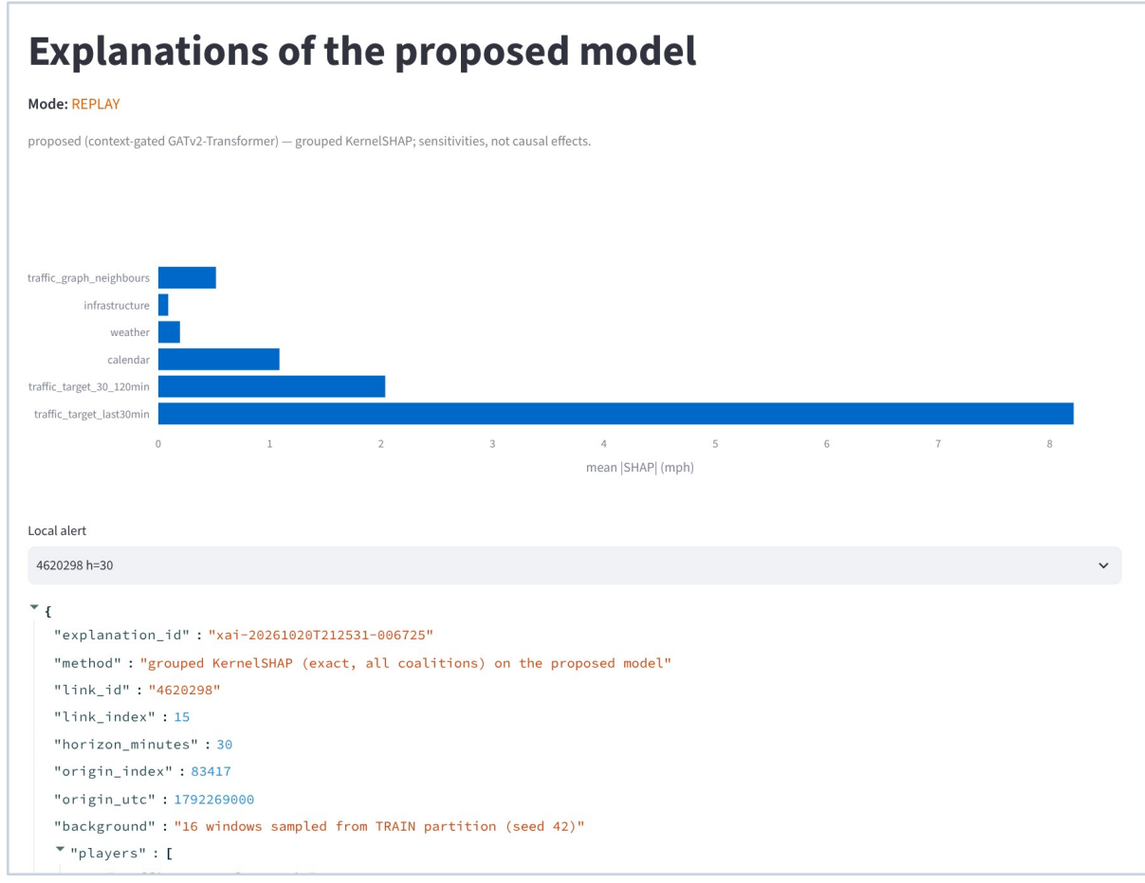
deferral_tradeoff.csv

missing_threshold	horizon_min	n_valid	retained	deferral_rate	mae_retained	mae_all	coverage_retained
0	15	144217	102928	0.2863	3.5605	4.7727	0.9531

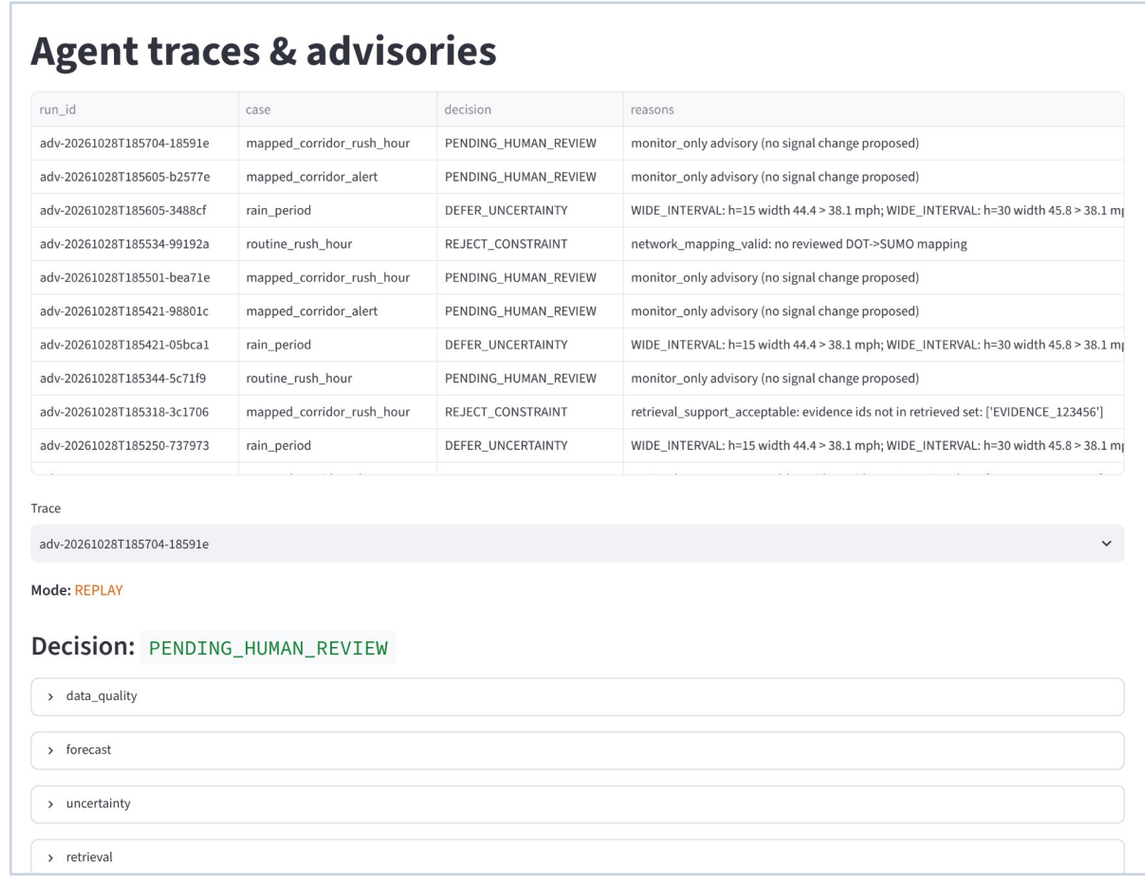
Screenshots of the project's Streamlit dashboard (src/smartcity/dashboard/app.py) captured on 28 Oct 2026; every number on them is read from committed result files. The LIVE page, run on 30 Oct 2026, is on slide 23. The Experiment results page shows the seed-42 runs; the three-seed results are on slide 22.

8. OUTCOME: WORKING DASHBOARD (4/5)

Explainability (proposed model) Mode shown: REPLAY



Agent trace & advisory Mode shown: REPLAY



Screenshots of the project's Streamlit dashboard (src/smartcity/dashboard/app.py) captured on 28 Oct 2026; every number on them is read from committed result files. The LIVE page, run on 30 Oct 2026, is on slide 23. Agent traces end in review, deferral or rejection; nothing is sent to real signals.

8. OUTCOME: WORKING DASHBOARD (5/5)

SUMO comparisons Mode shown: SIMULATION

SUMO comparisons

Mode: SIMULATION

Scenario-based microscopic simulation testbed using real-world road geometry (OSM) and declared demand assumptions. Not a calibrated digital twin; not real-world gains.

SUPERSEDED (C12) Matched-seed SUMO comparison of baseline vs protected controller (2026-10-20, 45 runs): 45 runs completed as recorded, but the 2026-10-21 testbed audit (simulation_testbed_audit.json) found they used bare from/to flows that SUMO re-routes at insertion (1,120 of 2,189 mainline routes left West Street) and a 0.2 m northbound entry stub, so they are not a corridor controller comparison; kept as evidence, no controller ranking claimed

Controller comparison (C44) Corrected controller comparison: 3 scenarios x 3 controllers x 5 matched seeds = 45 runs on the calibrated plan, all OK with 0 teleports; routine peak: fixed time and forecast rule identical (271.6 s mean time loss, 4,975 trips completed); workstation run 2026-10-28 18:15-18:40 UTC (re-run in full after the 26 Oct interruption at 39/45).

Controller comparison (C45) Event surge x1.25: forecast rule 426.0 s vs fixed time 450.2 s mean time loss (-5.4 %), completed 5,910 vs 5,720 of 6,712 loaded, unfinished 350 vs 540, mean corridor queue 180 vs 197 veh; incident: 443.9 vs 465.9 s (-4.7 %), 5,919 vs 5,725 completed, 341 vs 536 unfinished, 183 vs 203 queued; the direction holds in every seed (time-loss gain 12.9-29.9 s and 14.3-33.7 s; 148-250 more trips completed); seed means and per-seed differences computed from the CSV; SIMULATION on a calibrated scenario testbed, not a real-world gain.

Controller comparison (C46) The actuated controller runs on netconvert's actuated programs (signal_timing netconvert_guess, network 98e143b6f4e1296a), not on the calibrated plan, and is reported as an uncalibrated reference, not ranked; its time loss (384.8 / 538.2 / 554.4 s) is not compared with the matched pair.

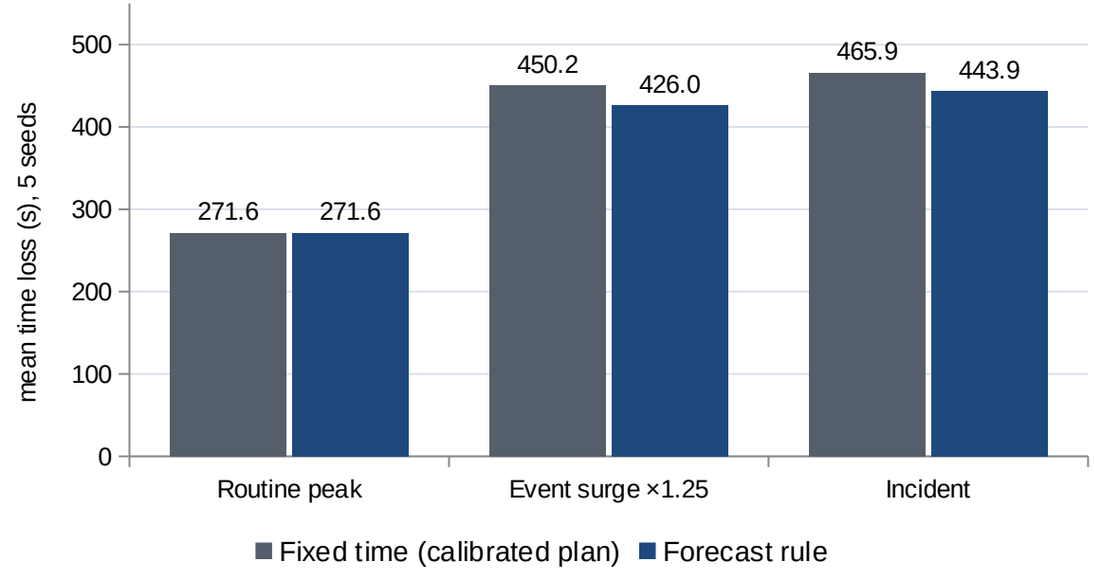
Testbed audit: preregistered demand-selection runs (fixed-time only)

v6_counts2025_cycle_splits_offsets (15 runs)

scenario	seed	main_nb_vph	main_sb_vph	side_vph_per_approach	loaded	completed_trips	unfinished_trips	teleports	corridor_mean_time_loss_s
sweep_side100	101	2510	2444	150	9443	7043	1735	76	652.0036
sweep_side100	102	2510	2444	150	9443	7063	1721	87	638.1816
sweep_side100	103	2510	2444	150	9443	7154	1628	85	630.1854

Captured 28 Oct 2026 from committed result files.

Corrected controller comparison Mode: SIMULATION



- 45 runs on the v6 calibrated plan, 5 matched seeds, 0 teleports (28 Oct 2026).
- Surge −5.4 %, incident −4.7 % time loss; 190–194 more of 6,712 trips finish; same direction in every seed. Routine peak: identical.
- **Simulated on a calibrated scenario testbed, not a real-world gain. The actuated controller is an uncalibrated reference and is not ranked.**

Source: reports/experiments/simulation_metrics.csv (claims C44–C46)

9. RESULTS & OUTCOMES

Held-out test period 6 Sep – 20 Oct 2026, 16 Manhattan links; advisory and retrieval results from the REPLAY workflow runs

Test MAE (mph): mean ± s.d. over seeds 42, 1 and 2

Model	15 min	30 min	60 min
Proposed GATv2–Transformer	**4.69 ± 0.06**	**5.34 ± 0.03**	**6.15 ± 0.02**
GRU	4.67 ± 0.06	5.37 ± 0.02	6.18 ± 0.01
Temporal Transformer	4.72 ± 0.03	5.38 ± 0.03	6.23 ± 0.08
Proposed, seed 42 only	4.77	5.33	6.15
Persistence	5.14	6.06	7.29

Paired day-block bootstrap (44 test days): the proposed model is lower than the temporal Transformer at every horizon (0.035 / 0.037 / 0.080 mph) and than the GRU at 30 and 60 min (0.032 / 0.031); the GRU is lower at 15 min (0.021). All intervals exclude zero, but with three seeds the advantage is small, not decisive.

Source: reports/experiments/multi_seed_summary.csv (nine GPU runs, 21 Oct 2026).

7–16 %
lower MAE than persistence (seed 42 run)

0.91 / 0.91 / 0.90
coverage of the 90 % conformal intervals; 0.76–0.80 under congestion

15–22 %
lower MAE on forecasts the reliability gate keeps (≈26 % deferred)

36 / 36
local-LLM responses schema-valid over three workflow passes (qwen2.5 7B)

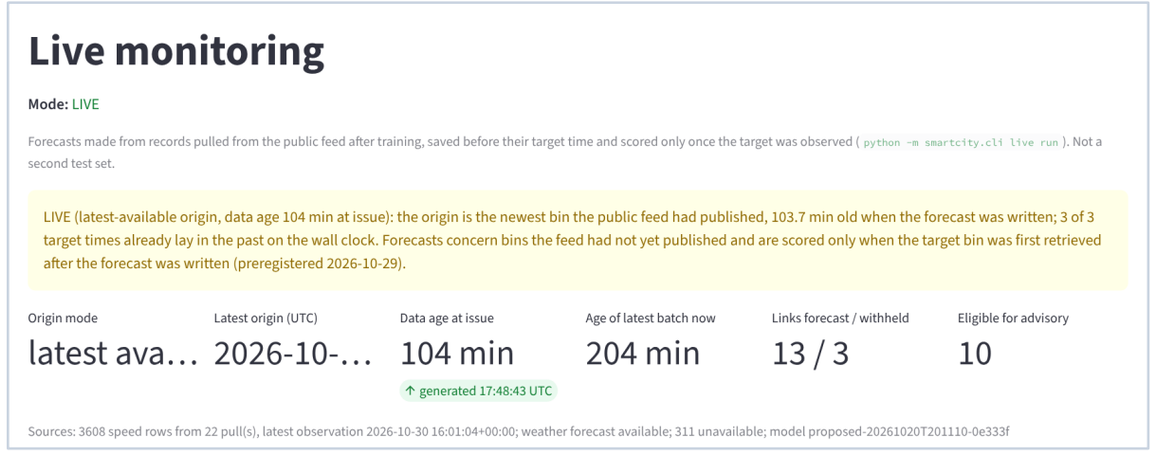
3 of 3 → 0 of 9
advisories citing non-existent evidence, without vs with retrieval

19
workflow runs per pass: 7 pending human review, 5 rejected by safety checks, 7 deferred

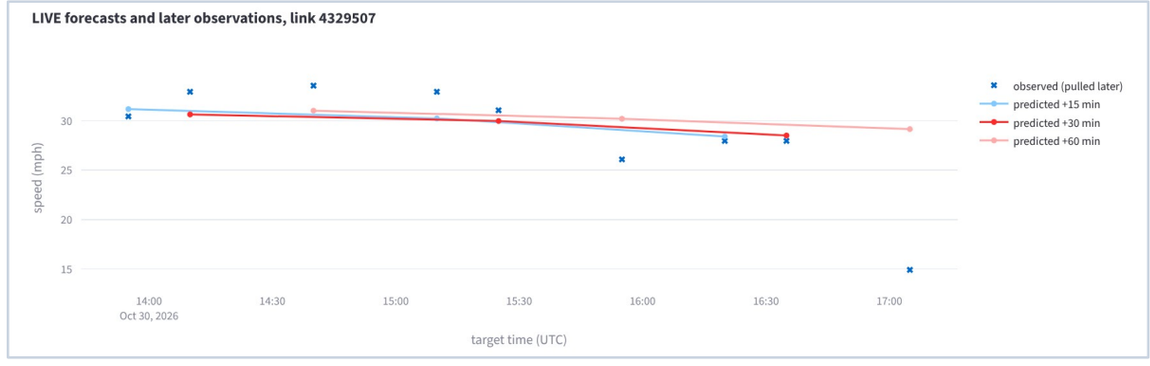
Also measured: removing the availability-aware gate raises MAE by 0.067 mph (95 % CI 0.048–0.086); removing operational weather features lowers it by 0.046 mph (a negative result). Simulated SUMO testbed: forecast rule −5.4 % / −4.7 % time loss (surge / incident) in all 5 seeds.

9. RESULTS & OUTCOMES: LIVE CHECK

Live monitoring Mode shown: LIVE



Forecasts vs later observations Mode shown: LIVE



Scored LIVE forecasts, 30 Oct 2026 Mode: LIVE

Horizon	MAE (mph)	Persistence	90 % coverage	Width (mph)
15 min	3.62	4.25	0.90	21.9
30 min	5.50	6.17	0.87	24.9
60 min	5.25	5.59	0.92	27.3
Eligible links		3.96 / 5.14 / 4.65		

- 117 forecasts scored (39 per horizon), 0 pending, 0 late; lower MAE than persistence at all three horizons. Eligible-links row: n = 30 per horizon.
- 3 origins: 13:40, 14:55, 16:05 UTC; data age at issue 148 / 89 / 104 min (mean 114); 13 of 16 links each, 3 withheld for missing recent records.

- **A forecast at the feed's own latency of bins it had not yet published, with its data age; not a real-time forecast.**
- **One afternoon, three origins, not a second test set.**
- **The wall-clock protocol stays NOT RUN: on 28 and 29 Oct the feed ran 1.7–4.3 h behind and every link was withheld.**

Source: reports/experiments/live_metrics.csv, artifacts/live/scored.csv (claims C50–C52)

9. RESULTS & OUTCOMES (CONTD.)

Implemented and verified

- Real-data pipeline on NYC DOT speeds (1 Jan – 20 Oct 2026), provenance hashed
- Seven forecasting models on identical targets; three-seed study of the deep models; ablations
- Conformal intervals, reliability gate, grouped SHAP
- FAISS retrieval (15 documents); local LLM with schema checks
- LangGraph workflow with 13 deterministic safety checks
- Calibrated SUMO testbed; 45-run controller comparison
- FastAPI, PostGIS jobs, Streamlit, MLflow, Docker Compose

Partial or not run

- SUMO gains are simulated on a calibrated scenario testbed, not NYC DOT timing; the first 45-run comparison is superseded
- LIVE: one afternoon at the feed's ~2 h latency; the wall-clock protocol stayed NOT RUN (feed 1.7–4.3 h behind)
- No real advisory has reached the simulation gate (the LLM chose monitor-only in all three passes)
- Only three seeds; the proposed model is not bit-reproducible on GPU at a fixed seed
- Road graph is sparse: 26 nodes, 16 edges
- Intervals under-cover congested periods

Outcomes and next steps

- IEEE conference paper (LaTeX and Word), every claim traced in the claims ledger
- GitHub repository with tests, runbooks and result files
- Next: repeat the LIVE run over more days; wall-clock forecasts need a fresher feed
- Next: more seeds without the compute budget
- Next: paper submission

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Thank You